

5.1 Challenges

Weak Spatial Reasoning and Representation. Multimodal LLMs (MLLMs) exhibit limited acuity in 3D spatial understanding, struggling with fine-grained relationships (e.g., front/back distinctions, occluded object localization) and precise geometric outputs (distances, angles). These issues stem partly from mismatches between unstructured point clouds and sequence-based LLM architectures, where high-dimensional 3D data incur prohibitive token counts or oversimplified encodings.

Data and Evaluation Gaps. Progress in 3D-aware LLMs is hindered by the scarcity of high-quality 3D-text paired datasets. Unlike the abundant resources for 2D images and video, the 3D domain lacks standardized, richly annotated datasets crucial for training robust models. Existing benchmarks focus mainly on discriminative tasks like classification and retrieval—emphasizing category differentiation rather than generating rich, descriptive 3D scene outputs. Consequently, evaluations often rely on subjective metrics (e.g., human or GPT-based judgments) that can lack consistency. Advancing the field requires developing objective, comprehensive benchmarks that assess both open-vocabulary generation and the spatial plausibility of descriptions relative to the underlying 3D structure.

Multimodal Integration and Generalization. Fusing 3D data (e.g., point clouds) with other modalities like 2D imagery, audio, or text poses significant challenges due to their distinct structural characteristics. The conversion and alignment of high-dimensional 3D data with lower-dimensional representations can lead to a loss of intricate details, diluting the original 3D richness. Moreover, current models often struggle with open-vocabulary recognition, limiting their ability to identify or describe objects outside of their training data—especially when encountering unseen scenes or novel objects. This difficulty is further compounded by the variability of natural language, from colloquial expressions to domain-specific terminology, and by noisy inputs. Thus, more sophisticated multimodal integration techniques and generalization strategies are needed to preserve geometric fidelity while accommodating diverse, unpredictable inputs.

Complex Task Definition. While 3D-aware LLMs excel in controlled settings, they lack frameworks for nuanced language-context inference in dynamic environments. Task decomposition and scalable encoding methods are needed to balance geometric fidelity with computational tractability, particularly for interactive applications requiring real-time spatial reasoning.

5.2 Future Directions

Enhancing 3D Perception and Representations. Addressing spatial reasoning gaps requires richer 3D-text datasets (e.g., from robotics, gaming, autonomous driving) and model architectures that encode geometric relationships. Multi-view data and robust depth cues can improve orientation, distance, and occlusion estimation. Compact 3D tokens and refined encoding/decoding methods may bridge unstructured point

clouds with sequence-based models, enabling fine-grained spatial understanding and generation.

Multi-Modal Fusion and Instruction Understanding. Tighter integration of modalities (point clouds, images, text, audio) via unified latent spaces or attention mechanisms could preserve subtle geometric and semantic details. Enhanced instruction processing—including hierarchical task decomposition, contextual interpretation, and robustness to dialects/terminology—would improve compositional reasoning in 3D environments and broaden real-world applicability. Furthermore, by leveraging these integrated representations, models can more adeptly adapt to complex instructions and novel scenarios, ultimately paving the way for more robust and versatile 3D reasoning systems.

Cross-Scene Generalization and Robust Evaluation. Open-vocabulary 3D understanding demands large-scale pretraining on diverse scenes and transfer/lifelong learning paradigms for adapting to novel objects or environments. This understanding extends beyond predefined categories to generalize to unseen objects and scenes. For instance, models need to comprehend “an old rocking chair” even if this specific type of chair never appeared in the training data.

Expanding Applications for Autonomous Systems. 3D-aware LLMs hold potential in robotics (navigation, manipulation), medical imaging (lesion detection), architectural design, and interactive education. Future systems may integrate environmental constraints, user perspectives, and object affordances for autonomous planning and decision-making in dynamic 3D contexts.

Collectively, these challenges and potential directions underscore the field’s rapid evolution and its equally significant open questions. Moving forward, more robust 3D-specific data resources, better model architectures, and more refined evaluation protocols will be essential to unlock the full potential of LLMs in three-dimensional settings—and ultimately bring intelligent, multimodal understanding closer to real-world deployment.

6 Conclusion

The integration of LLMs with 3D data is a dynamic research area. This survey categorized 3D-LLM research into image-based, point cloud-based, and hybrid modality-based spatial reasoning. It reviewed state-of-the-art methods, their applications in multiple fields, and associated challenges. Notably, image-based methods have data-related advantages but face issues like depth information shortage. Point cloud-based methods offer precise 3D details but encounter data-handling difficulties. Hybrid methods combine strengths yet struggle with data alignment. Applications are diverse, but challenges such as weak spatial perception, data scarcity, and evaluation problems exist. Future research should focus on enhancing 3D perception, improving multi-modal fusion, expanding generalization, developing evaluation metrics, enhancing instruction understanding, optimizing 3D representations, and exploring continuous learning. By addressing these, we can unlock the full potential of 3D-aware LLMs for real-world deployment and industry advancement.

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